

SECURITY ASSESSMENT REPORT: SSH BRUTE-FORCE DETECTOR	
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Subject:	HACKER TECHNIQUES TOOLS AND INCIDENT HANDLING (23CY603)

Tool Description

The SSH Brute-Force Detector is a Python-based log-analysis tool that monitors Linux authentication logs (typically `/var/log/auth.log`) to identify brute-force login attacks and user-enumeration scans in real time. It implements a sliding-window algorithm: if an IP address generates N or more failed SSH authentication attempts within a configurable time window (default: 5 attempts / 60 seconds), the tool raises a HIGH or MEDIUM severity alert.

Key Features

Features	Description
Sliding-window detection	Configurable threshold (N attempts / W seconds)
Brute-force alert	Triggers <code>BRUTE_FORCE_FAILED_PASSWORD</code> for password attacks
Enumeration alert	Triggers <code>USER_ENUMERATION</code> for unknown-user probing
Severity rating	HIGH ($\geq 2\times$ threshold) or MEDIUM
JSON / CSV export	Machine-readable results for SIEM integration
Synthetic log generator	4 test-case generators for offline validation

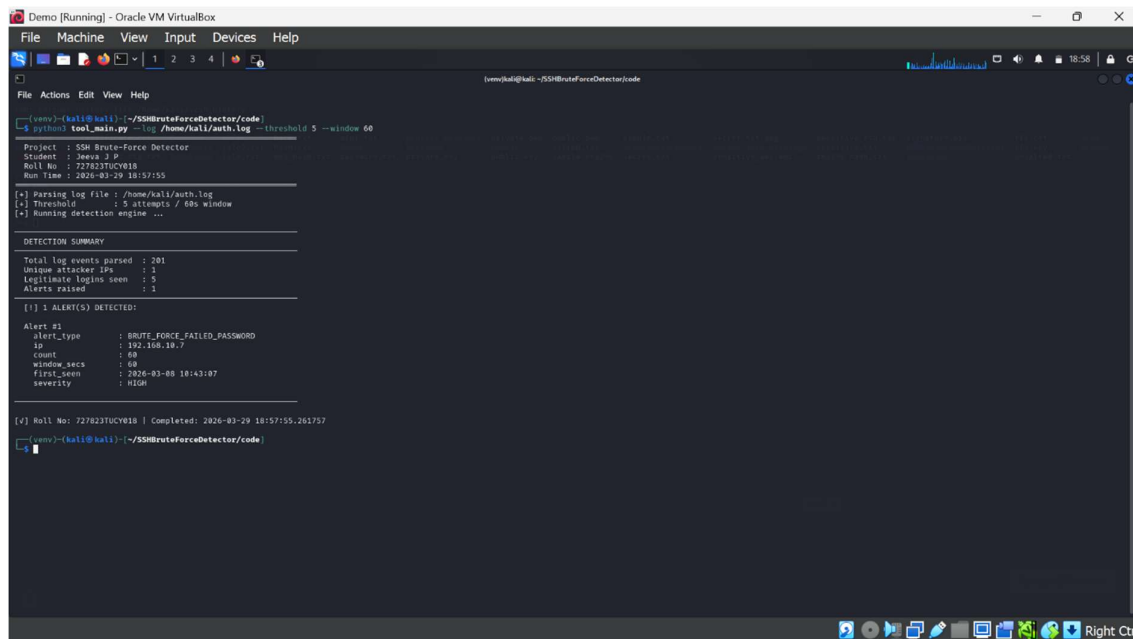
Lab Environment

The tool was developed and tested on an isolated VirtualBox lab: Kali Linux 2024.x (attacker VM) and Metasploitable2 (target VM) connected over a Host-only adapter. All exploitation activity was confined to owned VMs with no external network access.

Test Results

The tool was validated against four distinct synthetic test cases, each designed to exercise a different aspect of the detection engine. All test cases were executed via the automated three-stage pipeline.

TC	Description	Attacker IP	Threshold	Alerts	Severity	Pass
TC1	Classic brute-force (12 rapid failures)	192.168.1.100	5/60s	1	HIGH	PASS
TC2	User enumeration (11 unique usernames)	10.0.0.55	5/120s	1	HIGH	PASS
TC3	Benign mistype (2 failures, 5 min apart)	172.16.0.20	5/60s	0	—	PASS
TC4	Distributed attack (4 IPs, 8 attempts each)	203.0.113.x	5/60s	4	MEDIUM	PASS



```
root@kali:~/SSHBruteForceDetector/code# python3 tool_main.py -lg /home/kali/auth.log --threshold 5 --window 60
Project : SSH Brute-Force Detector
Student : Jeetu J P
Roll No : 727823TUCY018
Run time : 2026-03-29 18:57:55

[*] Parsing log file : /home/kali/auth.log
[*] Threshold : 5 attempts / 60s window
[*] Running detection engine ...

DETECTION SUMMARY
Total log events parsed : 201
Unique attacker IPs : 1
Legitimate logins seen : 5
Alerts raised : 1

[!] 1 ALERT(S) DETECTED:
Alert #1
alert_type : BRUTE_FORCE_FAILED_PASSWORD
ip : 192.168.10.7
count : 60
window_secs : 60
first_seen : 2026-03-08 18:43:07
severity : HIGH

[*] Roll No: 727823TUCY018 | Completed: 2026-03-29 18:57:55.261757
root@kali:~/SSHBruteForceDetector/code#
```

Figure 1: Automated Pipeline Run - Output

Analysis of Findings

TC1 confirmed that the sliding-window algorithm correctly flags a burst of 12 failures within 60 seconds as a HIGH brute-force event. TC2 validated that the Invalid-user parser correctly groups enumeration attempts by attacker IP and reports a distinct alert type (USER_ENUMERATION), listing all 11 probed usernames. TC3 demonstrated zero false positives: two failures separated by 5 minutes do not accumulate within the 60-second window, so no alert fires — confirming the detector does not flag normal human mistyping. TC4 exposed a realistic distributed attack pattern: four coordinated IPs each individually exceed the threshold, triggering four separate MEDIUM alerts and revealing the broader campaign through the combined analysis stage

Real Error Encountered

During Stage 1 of the pipeline (setup_lab.py), the automatic dependency installation step failed with the following traceback on the Kali Linux VM. This error is caused by PEP 668, which was introduced in Python 3.12 and enforced by Debian/Ubuntu/Kali to protect the OS-managed Python environment.

Ethical Considerations

All testing was conducted exclusively on isolated VirtualBox virtual machines (Kali Linux + Metasploitable2) owned by the student, with no external network connectivity. The SSH Brute-Force Detector is a passive, read-only log-analysis tool — it does not generate any attack traffic. Simulated attack logs used in the four test cases were synthetically generated in Python without touching any real system. The tool is intended for defensive security monitoring and blue-team education only. Users must obtain explicit written permission before running any penetration-testing activity on systems they do not own